

Problem Statements

Topics:

- Arrays
- Binary Search
- Programming Mathematics

The problems are designed to build strong fundamentals in problem solving, algorithmic thinking, and coding interview preparation.

Where an equivalent or closely related **LeetCode** problem is available, a direct link is provided.

1. Arrays

Sr. No.	Problem Title	Problem Statement	LeetCode Link
1	Sum of Array	Given an array of integers, calculate and return the sum of all elements in the array.	—
2	Sum of Even Elements	Given an array of integers, calculate the sum of all even elements present in the array.	—
3	Reverse Array	Given an array, reverse the order of its elements. The first element should become the last element and vice versa.	—
4	Move Zeros to End of List	Given an integer array, move all 0s to the end of the array while maintaining the relative order of the non-zero elements.	LeetCode 283 – Move Zeroes
5	Rotate Array by 1 Place	Given an array, rotate all elements one position to the right. The last element becomes the first element.	LeetCode 189 – Rotate Array
6	Check if Array is Sorted or Not	Given an array, determine whether the elements are arranged in non-decreasing order.	LeetCode 896 – Monotonic Array
7	Largest Element in the Array	Given an array of integers, find and return the largest element in the array.	—
8	Smallest Element in the Array	Given an array of integers, find and return the smallest element in the array.	—
9	Linear Search	Given an array and a target value, search for the target using linear search. Return its index	—

Sr. No.	Problem Title	Problem Statement	LeetCode Link
		if found; otherwise return -1.	
10	Count Occurrences	Given an array and a target value, count how many times the target element occurs in the array.	LeetCode 34 – Find First and Last Position
11	Find the Missing Number in Array of 1 to N	Given an array containing numbers from a specified range with one number missing, find the missing number.	LeetCode 268 – Missing Number
12	Running Sum of 1D Array	Given an array, create a new array where each element is the sum of all elements from the beginning of the array up to that position.	LeetCode 1480 – Running Sum of 1d Array

2. Binary Search

Binary Search is an efficient searching technique that works on a sorted or appropriately structured search space. It repeatedly divides the search space into two halves, giving an $O(\log n)$ search time in the standard case.

Sr. No.	Problem Title	Problem Statement	LeetCode Link
13	Basic Binary Search	Given a sorted array of integers and a target value, find the index of the target using Binary Search. Return -1 if the target is not present.	LeetCode 704 – Binary Search
14	Search Insert Position	Given a sorted array of distinct integers and a target value, return the index if the target exists. Otherwise, return the index where the target should be inserted to maintain sorted order.	LeetCode 35 – Search Insert Position
15	First Occurrence of Target	Given a sorted array that may contain duplicate values and a target, find the index of the first occurrence of the target. Return -1 if it is not present.	LeetCode 34 – Find First and Last Position
16	Last Occurrence of Target	Given a sorted array that may contain duplicate values and a target, find the index of the last occurrence of the target. Return -1 if it is not present.	LeetCode 34 – Find First and Last Position
17	First Missing Element in Sorted Array of 1 to N	Given a sorted array containing consecutive numbers with one or more missing values, use Binary Search to identify the first missing element.	LeetCode 268 – Missing Number

Sr. No.	Problem Title	Problem Statement	LeetCode Link
18	First Positive Element in Sorted Array	Given a sorted array containing negative, zero, and positive values, find the index of the first positive element using Binary Search.	—
19	First Element Greater Than or Equal to Target (Upper Bound)	Given a sorted array and a target value, find the first element whose value is greater than or equal to the target.	LeetCode 744 – Find Smallest Letter Greater Than Target
20	Lower Bound	Given a sorted array and a target value, find the first position where an element is greater than or equal to the target. If no such element exists, return the size of the array.	LeetCode 35 – Search Insert Position
21	Peak in Mountain Array	Given a mountain array that strictly increases and then strictly decreases, find the index of the peak element.	LeetCode 852 – Peak Index in a Mountain Array
22	Local Peak in Unsorted Array	Given an unsorted array, find an index of a peak element. A peak element is an element that is greater than its neighboring elements.	LeetCode 162 – Find Peak Element
23	Minimum in Rotated Sorted Array	Given a sorted array that has been rotated an unknown number of times, find the minimum element using Binary Search.	LeetCode 153 – Find Minimum in Rotated Sorted Array
24	Search in Rotated Sorted Array	Given a rotated sorted array and a target value, search for the target and return its index. Return -1 if the target does not exist.	LeetCode 33 – Search in Rotated Sorted Array

Note: In standard Binary Search terminology, **Lower Bound** is the first position containing a value $\geq \text{target}$. The LeetCode Search Insert Position problem implements this same first-valid-position pattern.

3. Programming Maths

Sr. No.	Problem Title	Problem Statement	LeetCode Link
25	Sum of Digits	Given a non-negative integer, repeatedly add its digits until a single digit remains.	LeetCode 258 – Add Digits
26	Reverse Integer	Given a signed 32-bit integer, reverse its digits. If the reversed integer falls outside the signed 32-bit integer range, return 0.	LeetCode 7 – Reverse Integer

Sr. No.	Problem Title	Problem Statement	LeetCode Link
27	Palindrome Number	Given an integer, determine whether the number reads the same forward and backward.	LeetCode 9 – Palindrome Number
28	Count Trailing 0s in Factorial of N	Given an integer n , determine the number of trailing zeroes in $n!$.	LeetCode 172 – Factorial Trailing Zeroes
29	Happy Number – Slow Pointer & Fast Pointer	Determine whether a number is a happy number. Use the slow and fast pointer technique to detect a cycle in the sequence generated by repeatedly replacing the number with the sum of the squares of its digits.	LeetCode 202 – Happy Number
30	Print Factors of a Number – $O(N)$	Given a positive integer N , print all factors/divisors of N by checking every number from 1 to N .	—
31	Print Factors of a Number – $O(\sqrt{N})$	Given a positive integer N , print all factors/divisors efficiently by checking only up to $\sqrt{N}$. For every divisor found, its corresponding paired divisor should also be considered.	—
32	Check if Number is Perfect	Given a positive integer, determine whether it is a perfect number. A perfect number is equal to the sum of its positive divisors excluding itself.	LeetCode 507 – Perfect Number
33	Count Factors – $O(\sqrt{N})$	Given a positive integer N , count the total number of factors of N using an $O(\sqrt{N})$ approach.	—
34	Check if Number is Prime – $O(\sqrt{N})$	Given a positive integer N , determine whether it is a prime number. Use divisibility checks only up to $\sqrt{N}$.	—
35	Print All Prime Numbers from 1 to N – $O(N\sqrt{N})$	Given an integer N , find and print all prime numbers between 1 and N by checking the primality of each number.	LeetCode 204 – Count Primes
36	Print All Prime Numbers – Sieve of Eratosthenes	Given an integer N , find all prime numbers less than or equal to N using the Sieve of Eratosthenes algorithm.	LeetCode 204 – Count Primes

LeetCode Problems Used

The following LeetCode problems are directly or closely related to the exercises in this repository:

Arrays

1. [Move Zeroes – LeetCode 283](#)
2. [Rotate Array – LeetCode 189](#)
3. [Monotonic Array – LeetCode 896](#)
4. [Find First and Last Position – LeetCode 34](#)
5. [Missing Number – LeetCode 268](#)
6. [Running Sum of 1d Array – LeetCode 1480](#)

Binary Search

7. [Binary Search – LeetCode 704](#)
8. [Search Insert Position – LeetCode 35](#)
9. [Find First and Last Position – LeetCode 34](#)
10. [Find Smallest Letter Greater Than Target – LeetCode 744](#)
11. [Peak Index in a Mountain Array – LeetCode 852](#)
12. [Find Peak Element – LeetCode 162](#)
13. [Find Minimum in Rotated Sorted Array – LeetCode 153](#)
14. [Search in Rotated Sorted Array – LeetCode 33](#)

Programming Maths

15. [Add Digits – LeetCode 258](#)
16. [Reverse Integer – LeetCode 7](#)
17. [Palindrome Number – LeetCode 9](#)
18. [Factorial Trailing Zeroes – LeetCode 172](#)
19. [Happy Number – LeetCode 202](#)
20. [Perfect Number – LeetCode 507](#)
21. [Count Primes – LeetCode 204](#)

Learning Objectives

By completing these problems, students will practice:

- Array traversal
- Array manipulation
- Searching techniques

- Prefix sums
 - Linear Search
 - Binary Search
 - Lower Bound and Upper Bound
 - First and Last Occurrence
 - Rotated Sorted Arrays
 - Peak Element problems
 - Number manipulation
 - Digit extraction
 - Factorization
 - Prime number detection
 - Time complexity analysis
 - Sieve of Eratosthenes
 - Slow and Fast Pointer technique
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